

Introduction to Algebra — Course Syllabus

Mary, Star of the Sea School · SY 2026-2027

Mary, Star of the Sea School Mission

Nurture, Unite, Inspire. We are committed to nurturing each child's curiosity, uniting our classroom in respect and collaboration, and inspiring every student to grow in knowledge, faith, and wonder for the world around them.

Course Description

Math 8 is a full high-school Algebra 1 course. You will work with linear functions, systems of equations, exponents and scientific notation, the Pythagorean theorem, and exponential and inverse-variation relationships — all the algebra needed for a confident start in high-school math. The course is paced to support STEM class throughout the year: chemistry uses linear equations and weighted averages, motion labs use physics equations of motion and trigonometry, and the spring electricity-and-waves unit relies on Ohm's Law, inverse variation, and Kepler's geometry. Class time balances skill-building practice, project-based summatives, lab work, and direct connection to real Hawaiian and engineering problems.

Textbook: Glencoe Math Algebra 1

Teacher

Dr. Peter Park

Email: ppark@starofthesea.org

Phone: (808) 228-2285

Parents may email me or leave a message with the school. I respond within 24 hours.

Materials

- Composition notebook
 - Pencils, erasers, highlighters
 - Notebook paper
 - Calculator
 - Folder for handouts
 - Textbook (Glencoe Math Algebra 1)
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Grading Policy

20%	25%	30%	25%
Participation	Tests	Homework	Quizzes
Composition books, notes, in-class behavior, supplies	Q&A, written essay (1 per chapter, 20 pts)	Assignments, classwork, class prep, projects	2 per chapter, 5 pts each

Make-ups: Students who score below 80% on tests may make corrections or retest to improve their grade.

Late work: Homework is due the next class meeting. Late work earns a maximum of 50% credit; if not turned in after two class meetings, the grade is a zero.

Course Outline by Quarter — Introduction to Algebra

Q1 — Lab Math & Number Foundation

- Lab Math Refresher: Significant Figures, Percent Error, Scientific Notation with computation, Unit Conversions, Density review
- Real Numbers; Square Roots
- Exponents & Laws of Exponents

Q2 — Lines, Slopes & Stoichiometry

- Slope & Rate of Change

- Linear Equations (Standard, Slope-Intercept)
- Rearranging Formulas (essential for stoichiometry)
- Systems of Equations — ready for STEM chemistry unit
- Percent & Weighted Averages — for the STEM Element Babybook (isotope abundance) and Stain Lab (concentrations)

Q3 — Math for Motion (Front-Loaded)

- Physics Equations of Motion — taught early January for STEM Velocity / Acceleration / Projectile Labs
- Functions; Linear Functions
- Pythagorean Theorem; Intro to Trigonometry (SOH-CAH-TOA)
- Transformations; Angle Relationships

Q4 — Electricity, Waves & Cosmos

- Ohm's Law & Linear Functions; Series and Parallel Circuits
- Waves — Frequency & Wavelength (inverse variation)
- Optics & Snell's Law (uses Q3 trig)
- Kepler's Laws of Orbital Mechanics
- Volume; Scatter Plots; Two-Way Tables; Word Problems capstone

Student & Parent Communication

Beehively is the platform where students and parents can regularly check grade progress and receive notifications of missing or incomplete assignments.

Parents are contacted by email up to two times if their child is receiving a deficiency, plus standard email notifications. Students are always informed of any grade that is a C or lower and ultimately are responsible for their grades.

Student work is updated regularly.

Middle School Math Goal

From 6th through 8th grade, we focus on more than just numbers — we work to build strong math skills, boost confidence, and develop critical problem-solving abilities that students can apply in every subject and in everyday life. Our approach combines hands-on activities, real-world applications, and a supportive learning environment where students feel encouraged to take risks, ask questions, and think creatively.

By the time they graduate, our students are not only ready for Algebra and beyond, but they also possess the perseverance, analytical thinking, and resourcefulness they need to thrive in high school and in life.